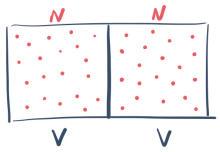
Physics 4B Spring 2026
Notes on the Gibbs Paradox

TA: Jiashu Han
June 4, 2026

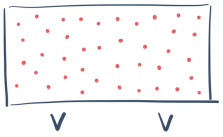
1. **The Paradox**. The Gibbs paradox considers the following scenario:

   - A box is separated into two equal halves of volume V by a partition. The gases in the two halves are identical and have the same number of particles N, pressure P and temperature T.

   - The partition is removed. Because the gases are identical, the initial and final equilibrium states are thermodynamically indistinguishable, thus the entropy change should be zero[1].

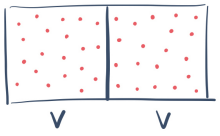
     The gas in each half expands to twice its original volume, while maintaining the same temperature.

     For an isothermal process, the entropy change is

     $$\Delta S = \Delta S_{\text{left}} + \Delta S_{\text{right}} = 2Nk_B \ln\left(\frac{2V}{V}\right) = 2Nk_B \ln 2 > 0. \tag{1}$$

   - We reinsert the partition, which should return the system to the original state. Since entropy is a state variable, if (1) were correct, it would imply a decrease of entropy, which violates the second law of thermodynamics and thus creating a paradox.

[1] Even though the process of removing the partition in general isn't reversible.

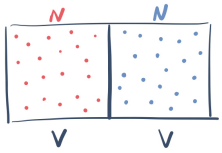
Short answer: The paradox is the result of counting the number of states of the gases without considering the indistinguishability of the gas particles. The naive calculation of the entropy change in (1) did not take into account the indistinguishability of the particles.

2. **A quick note on intensive and extensive quantities**. If you look up the Gibbs paradox, you may quickly find discussions about issues with entropy not being an extensive property of a thermodynamic system. While this is at the heart of what entropy is, we will not focus on this aspect in our discussion here. Essentially, an extensive property is one that is proportional to the amount of substance in the system. Thermodynamic entropy $S = k_B \ln \Omega$ is additive for two independent systems A and B:

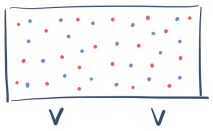
$$S_{AB} = k_B \ln(\Omega_A \Omega_B) = k_B \ln \Omega_A + k_B \ln \Omega_B = S_A + S_B, \tag{2}$$

and with the correct counting of indistinguishable particles it is also extensive (which is exactly what the following discussion focuses on).

3. **Explanation and Resolution**. If the gas particles are distinguishable (e.g. hydrogen in the left half and oxygen in the right half),



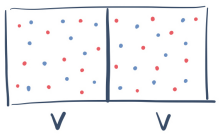
mixing them would result in an increase in entropy as the end result is both gases evenly distributed in a larger volume:



Treating this as an isothermal process (since each half of the gas undergoes free-expansion, which results in a new state on the same isotherm as the old state), the change in the thermodynamic entropy for each half is

$$\Delta S = \int_i^f \frac{dQ_{\text{rev}}}{T} = \int_V^{2V} \frac{N k_B \, dV'}{V'} = N k_B \ln 2 \tag{3}$$

There's actually no paradox here, because the two gases are even mixed now, and putting the partition back would not restore the old state,



and there would not be a violation of the second law.

We just explained this scenario from the macroscopic perspective. Now let's look at the microscopic. For a simple illustration we will ignore the momentum of the particles and constants in the counting of microstates. The number of possible locations each particle can take is proportional to the volume V. Naively, there are $\propto V^N$ ways to place the N particles, but we need to divide this by $N!$ if all of these N particles are indistinguishable from one another, because changing their orders won't make a difference in the overall configuration, and there are $N!$ ways to order them.

Before the partition was removed, the number of possible microstates for the whole system is just the product of the number of possible microstates for the halves

$$\Omega_1 \propto \left(\frac{V^N}{N!}\right)\left(\frac{V^N}{N!}\right) = \frac{V^{2N}}{(N!)^2} \tag{4}$$

After removing the partition, each half spreads to fill the whole box, and the number of possible microstates is the product of the numbers of possible microstates for the expanded halves:

$$\Omega_2 \propto \left(\frac{(2V)^N}{N!}\right)\left(\frac{(2V)^N}{N!}\right) = \frac{(2V)^{2N}}{(N!)^2} \tag{5}$$

The change in entropy is

$$\Delta S_{12} = k_B \ln\left(\frac{\Omega_2}{\Omega_1}\right) = 2Nk_B \ln 2 > 0 \tag{6}$$

as expected.

After re-inserting the partition, there should be $N/2$ hydrogen molecules and $N/2$ oxygen molecules in each half of the box, meaning that for each half, the number of possible microstates follows the same calculation as in Ω_2 but with everything scaled down by a factor of 2:

$$\Omega_{3,\text{half}} \propto \left(\frac{V^{N/2}}{(N/2)!}\right)\left(\frac{V^{N/2}}{(N/2)!}\right) = \frac{V^N}{((N/2)!)^2} \tag{7}$$

so the number of possible microstates is

$$\Omega_3 = \Omega_{3,\text{half}}^2 \propto \frac{V^{2N}}{((N/2)!)^4} \tag{8}$$

Using Stirling's approximation

$$\ln N! \overset{\text{large}N}{\to} N \ln N - N + O(\ln N) \tag{9}$$

we can show that the change in entropy is

$$\begin{aligned}
\Delta S_{23} &= k_B \ln\left(\frac{\Omega_3}{\Omega_2}\right) = k_B \ln\left(\frac{(N!)^2}{((N/2)!)^4 2^{2N}}\right) \\
&\approx k_B \left(2N \ln N - 2N - 4(N/2)\ln(N/2) + 2N - 2N \ln 2\right) \\
&= k_B \left(2N \ln N - 2N \ln(N/2) - 2N \ln 2\right) \\
&= 0
\end{aligned} \tag{10}$$

which is exactly what we expected (since nothing really changed from state 2 to state 3), and it does not violate the second law of thermodynamics. Again, as a reminder, there's no paradox in this scenario, and we are getting exactly what we expected here, because once the two gases mix, we can't return the system automatically to the initial state by simply putting back the partition.

The paradox appears when the gas particles in the two halves are identical / indistinguishable (e.g. hydrogen in both halves). Mixing the two halves should make no difference as the state variables (P, V, T) all remain well defined and unchanged, and the process of removing the partition should lead to no change in entropy. Despite this, the calculation in (3) seems to still apply to our case, which means we would still get

$$\Delta S_{12} = 2N k_B \ln 2 \tag{11}$$

But is this correct? Let's count the microstates again. Before the partition was removed, the number of microstates is the same as before, but divided by a factor of 2 because gas A and gas B are identical and we don't care if A is on the left or B is on the left - they count as not two but one single microstate[2]:

$$\Omega_1 \propto \frac{1}{2}\left(\frac{V^N}{N!}\right)\left(\frac{V^N}{N!}\right) = \frac{V^{2N}}{2(N!)^2} \tag{12}$$

What's different from the previous case (distinguishable particles) is is after the partition is removed. Since the particles from the two halves are indistinguishable,

[2]This factor of 1/2 here is somewhat unnecessary, especially since in the previous scenario we did not actually count the case where gas A is in the right and gas B is in the left, but it doesn't actually change the result so I choose to keep it here to highlight the indistinguishability. If we removing this factor, then there's no need to drop the term in (14).

we cannot simply multiply the number of microstates from the halves together. Instead, the multiplicity factor on the bottom would be $(N+N)! = (2N)!$

$$\Omega_2 \propto \frac{(2V)^{2N}}{(2N)!} \tag{13}$$

The change in entropy is[3]

$$\begin{aligned}\Delta S_{12} &= k_B \ln\left(\frac{\Omega_2}{\Omega_1}\right) = k_B \ln\left(\frac{2^{2N+1}(N!)^2}{(2N)!}\right) \approx k_B \ln\left(\frac{2^{2N}(N!)^2}{(2N)!}\right) \\ &\approx k_B \left(2N\ln 2 + 2N\ln N - 2N - 2N\ln(2N) + 2N\right) \\ &= 0\end{aligned} \tag{14}$$

This is where the paradox appears - the previous thermodynamic calculation in (3) treats the gases on the two sides as distinct components, but they are in fact identical, so the number of microstates is overestimated by a factor of $N!$ if we assume distinguishability (compare (5) and (13)). In (13) we already took into account the indistinguishability of the particles, which led to the correct cancelation of the $2Nk_B \ln 2$ term in ΔS_{12}.

The rest of the procedure remains the same: after restoring the partition, the number of microstates is the same as before the partition is removed (12)

$$\Omega_3 = \Omega_1 \tag{15}$$

and the change in entropy is $\Delta S_{23} = 0$.

We have thus resolved the paradox. Essentially, the answer is that (3) is the wrong formula to use in the case of identical particles. The Gibbs paradox shows that entropy must be calculated with careful consideration of the distinguishability of the particles. It's really a question about what constitutes a distinct species and which permutations correspond to physically new states. As long as the calculation of entropy correctly takes into account the distinguishability of the particles, there will be no paradox.

4. **What does distinguishability mean exactly?** It turns put the Gibbs paradox is all about correctly addressing the distinguishability of particles, so what do mean by distinguishable exactly? For example, in the above discussion, we have gas A (hydrogen) in the left and gas B (oxygen) in the right. If you thought about it already, you would realize that we treated gas A particles as indistinguishable

[3]Note that we approximated the exponent $2N+1$ in the first line as $2N$. This is totally fine because this extra 1 contributes only an $O(1)$ term in the result, whereas we are already dropping all terms smaller than $O(\ln N)$ by doing the Stirling approximation.

among themselves (and also the gas B particles among themselves), but gas A particles are distinguishable from the gas B particles.

That's exactly what we did. When we say two gases are "different," we do not mean that every particle gets a unique label. Instead, we mean:

- All A particles are identical to one another.
- All B particles are identical to one another.
- An A particle can be distinguished from a B particle.

Thus when A and B are mixed together after the partition is removed, the correct counting in (5) is $\Omega \propto \frac{1}{N_A!}\frac{1}{N_B!}$, not $\Omega \propto \frac{1}{(N_A+N_B)!}$ as in (13).

5. **How do we know the particles are distinguishable?** You may also ask: how do we know a priori whether the particles are distinguishable or indistinguishable? What determines their distinguishability? Is it the (macroscopic) physical laws they follow, or the microscopic properties of the particles?

   Consider the air molecules, which constitutes many species (N_2, O_2, etc.). Each of these species behave like an ideal gas, which means they follow exactly the same equation of state ($PV = Nk_BT$) and have the same thermodynamic behavior. It is intriguing to think that they should be considered identical because they follow the same macroscopic laws.

   On the other hand, these species are technically distinguishable from one another because they are different molecules with different masses, different rotational spectra, different chemical properties, and different scattering cross sections. In principle, we can measure whether a given molecule is N_2 or O_2[4].

   The above question gets right to the heart of what "distinguishable" means in statistical mechanics. The short answer is that the microscopic properties determine distinguishability. Even if two species obey the same equation of state (macroscopic behavior), they are still treated as distinguishable if they are physically different species in the microscopic scale.

6. **An exercise on counting**. To provide you with more context, we will use the coins in box problem from the discussion worksheet as an example.

   - Say we have a box with a total of $2N$ slots, each of which can hold exactly one coin. We shake the box constantly so the coins can flip sides or exchange position with its neighbors. Initially the box is partitioned into two halves

[4]Microscopic particles of the same species are truly indistinguishable from one another, this is known a priori by postulates of quantum mechanics.

each with N slots, and each half has $M < N$ coins. The partition is then removed, and the M coins from each half are free to move around the entire box. What is the entropy of the system before and after the partition is removed?

Let's consider the problem in 3 different cases:

(a) All of the coins are exactly identical.

(b) The two halves hold two distinct species of coins (left half has A, and the right half has B).

(c) Every coin is unique and can be distinguished from others.

For case (a), the number of microstates before the partition is removed (or after it's restored) would be[5]

$$\Omega_1 = \Omega_3 = \frac{1}{2}\binom{N}{M}^2 2^{2M} \tag{16}$$

After the partition is removed, the total number of microstates would be

$$\Omega_2 = \binom{2N}{2M} 2^{2M} \tag{17}$$

The change in entropy for removing the partition would be

$$\begin{aligned}
\Delta S_{12} &= k_B \ln\left(\frac{2\binom{2N}{2M}}{\binom{N}{M}^2}\right) = k_B \ln\left(\frac{2(2N)!(M!)^2((N-M)!)^2}{(2M)!(2N-2M)!(N!)^2}\right) \\
&\approx k_B[\ln 2 + 2N\ln(2N) - 2N + 2M\ln M - 2M \\
&\quad +2(N-M)\ln(N-M) - 2(N-M) - 2M\ln(2M) + 2M \\
&\quad -(2N-2M)\ln(2N-2M) + (2N-2M) - 2N\ln N + 2N] \\
&= k_B[\ln 2 + 2N\ln 2 - 2M\ln 2 - (2N-2M)\ln 2] \\
&\approx 0
\end{aligned} \tag{18}$$

where we dropped all terms smaller than $O(\ln N)$, $O(\ln M)$, or $O(\ln(N-M))$.

The change of entropy for putting back the partition is $\Delta S_{23} = k_B \ln(\Omega_3/\Omega_2) = 0$.

For case (b), the number of microstates before the partition is removed is

$$\Omega_1 = \binom{N}{M}^2 2^{2M} \tag{19}$$

[5]Again there's the extra factor of 1/2 I choose to keep, but it's totally ok to drop it.

The number of microstates after the partition is removed is

$$\Omega_2 = \binom{2N}{M}^2 2^{2M} \tag{20}$$

and the change in entropy is

$$\begin{aligned}
\Delta S_{12} &= k_B \ln \left(\frac{\binom{2N}{M}^2}{\binom{N}{M}^2} \right) = 2k_B \ln \left(\frac{(2N)!M!(N-M)!}{M!(2N-M)!N!} \right) \\
&\approx 2k_B[2N\ln(2N) - 2N + (N-M)\ln(N-M) - (N-M) \\
&\quad -(2N-M)\ln(2N-M) + (2N-M) - N\ln N + N] \\
&\approx 2k_B[N\ln N + 2N\ln 2 + (N-M)\ln(N-M) \\
&\quad -(2N-M)\ln(2N-M)]
\end{aligned} \tag{21}$$

It's not easy to tell whether or not this is non-negative, but we can look at some limits. For $N = M$, we would get $\Delta S_{12} = 4Nk_B \ln 2 > 0$. For $N \gg M$, we would get $\Delta S_{12} = 0$. Both make sense and do not violate the second law.

Following the same idea in (7) and (8), the number of microstates after the partition is restored would be

$$\Omega_3 = \binom{N}{M/2}^4 2^{2M} \tag{22}$$

and the change in entropy for restoring the partition would be

$$\begin{aligned}
\Delta S_{23} &= k_B \ln \left(\frac{\binom{N}{M/2}^4}{\binom{2N}{M}^2} \right) = 2k_B \ln \left(\frac{(N!)^2 M!(2N-M)!}{((M/2)!)^2((N-M/2)!)^2(2N)!} \right) \\
&\approx 2k_B[2N\ln N - 2N + M\ln M - M + (2N-M)\ln(2N-M) \\
&\quad -(2N-M) - M\ln(M/2) + M - 2(N-M/2)\ln(N-M/2) \\
&\quad +(2N-M) - 2N\ln(2N) + 2N] \\
&= 2k_B \left[(M-2N)\ln 2 + (2N-M)\ln\left(\frac{2N-M}{N-M/2} \right) \right] \\
&= 0
\end{aligned} \tag{23}$$

as expected.

Finally, for case (c), the number of microstates before the partition is removed would be

$$\Omega_1 = \left(\frac{N!}{(N-M)!}\right)^2 2^{2M} \tag{24}$$

with no $1/N!$ factors this time because the coins are individually distinguishable. After the partition is removed, the number of microstates would be

$$\Omega_2 = \left(\frac{(2N)!}{(2N-2M)!}\right) 2^{2M} \tag{25}$$

and the change in entropy for removing the partition would be

$$\begin{aligned}
\Delta S_{12} &= k_B \ln\left(\frac{(2N)!((N-M)!)^2}{(2N-2M)!(N!)^2}\right) \\
&\approx k_B[2N\ln(2N) - 2N + 2(N-M)\ln(N-M) - 2(N-M) \\
&\quad -(2N-2M)\ln(2N-2M) + (2N-2M) - 2N\ln N + 2N] \\
&= k_B[2N\ln 2 - 2(N-M)\ln 2] \\
&= 2Mk_B \ln 2
\end{aligned} \tag{26}$$

After restoring the partition, you would be tempted to think that the number of microstates would be the same as Ω_1, but there's a difference. In state 1, coins labeled 1 through M are restricted to be in the left half, while the coins labeled $M+1$ through $2M$ are restricted to be in the right half. Now these restrictions are gone, so there are more possibilities - we must count how many ways we can choose half of the coins to place in one particular half of the box

$$\Omega_3 = \binom{2M}{M}\left(\frac{N!}{(N-M)!}\right)^2 2^{2M} \tag{27}$$

The change in entropy for restoring the partition is therefore

$$\begin{aligned}
\Delta S_{23} &= k_B \ln\left(\frac{\binom{2M}{M}\left(\frac{N!}{(N-M)!}\right)^2}{\frac{(2N)!}{(2N-2M)!}}\right) = k_B \ln\left(\frac{(2M)!(N!)^2(2N-2M)!}{(M!)^2((N-M)!)^2(2N)!}\right) \\
&\approx k_B\ [2M\ln(2M) - 2M + 2N\ln N - 2N + (2N-2M)\ln(2N-2M) \\
&\quad -(2N-2M) - 2M\ln M + 2M - 2(N-M)\ln(N-M) \\
&\quad +2(N-M) - 2N\ln(2N) + 2N] \\
&= k_B\ [2M\ln 2 + 2(N-M)\ln 2 - 2N\ln 2] \\
&= 0
\end{aligned} \tag{28}$$

Once again, this result obeys the second law of thermodynamics!